

Motorola M6800 Support Products

For Total Microcomputer System Design

The evolution of large-scale integration (LSI) has spawned the microprocessor as a low-cost circuit component, and has created a viable alternative to the conventional hardwired method of circuit, and system design. The resultant concept of utilizing a programmable microcomputer to perform control functions previously relegated to dedicated, single-function circuitry presents such a powerful approach to the implementation of control systems that it is widely acclaimed as the single most important development of the past decade.

To facilitate the design of pervasive and cost-effective microcomputer-based systems, Motorola has generated a repertoire of basic LSI building blocks. Appropriate combination of these components permits the establishment of an end-system with any desired degree of sophistication. Known as the M6800 Microcomputer Family, this repertoire at present includes the following components:

MC6800 — A Microprocessing Unit (MPU) which represents the heart of a microcomputer system. Encompassed within this component are the arithmetic and control capabilities that are fundamental to any computer system.

MC6820 — A Peripheral Interface Adapter (PIA) that interfaces the MPU with a variety of input/output peripherals such as terminals, printers, etc. Its parallel I/O architecture interfaces with a parallel oriented peripheral.

MC6850 — An Asynchronous Communications Interface Adapter (ACIA) for communications between the MPU and a serially oriented peripheral, such as a Modem. This converts serially transmitted data to the parallel orientation required by the MPU, and vice versa.

MC6852 — A Synchronous Serial Data Adapter (SSDA) for communications between the MPU and a bit-serial peripheral.

MC6860 — A low-speed Modem (modulator-demodulator) that permits the MPU to be accessed, through the ACIA, by data transmitted over the telephone lines.

MCM6830 — A 1024 x 8-bit Read-Only Memory (ROM), programmed by the factory with the appropriate instructions stipulated by the designer's end-system requirements.

MCM6810 — A 128 x 8-bit Random Access Memory (RAM) with access time as low as 350 ns (MC6810AL1).

All of the above parts are mutually compatible in a bus-oriented system that requires no additional buffers or other outboard components. An advanced N-channel MOS technology yields reliable operation while a powerful 72-statement instruction set provides a high degree of programming flexibility.

But successful microcomputer-based system design requires more than a set of compatible "hardware" components. Even after a "working computer" has been implemented through the interconnection of the required components, there still remains the development and verification of the "program" that converts a mindless computer into a dedicated functional system. The M6800 system, therefore, comprises an extensive and growing array of support products dedicated to insuring the timely and successful completion of a design effort.

The Support Products

Motorola's total M6800 System Design Support concept encompasses the following categories:

Support Hardware — A series of instruments that emulate, demonstrate and evaluate the operation of any M6800-based design.

Support Peripherals — A growing array of peripheral equipment that interfaces with M6800 system designs to increase data throughput during system development.

Support Software — Development and diagnostic programs that permit the design of end-use programs utilizing Time-Share computers, in-house computers or the support hardware mentioned above.

Design Support — Includes documentation, training, applications assistance and a staff of professionals to answer questions and help solve system development problems.

The information to follow describes, in detail, these support products.



MOTOROLA MICROSYSTEMS

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